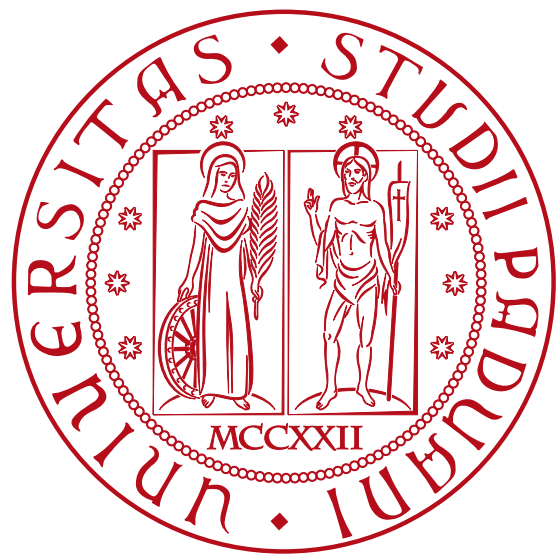




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Gianmaria Silvello

Full Professor @ Università degli Studi di
Padova | Data Integration, Semantic Mod...

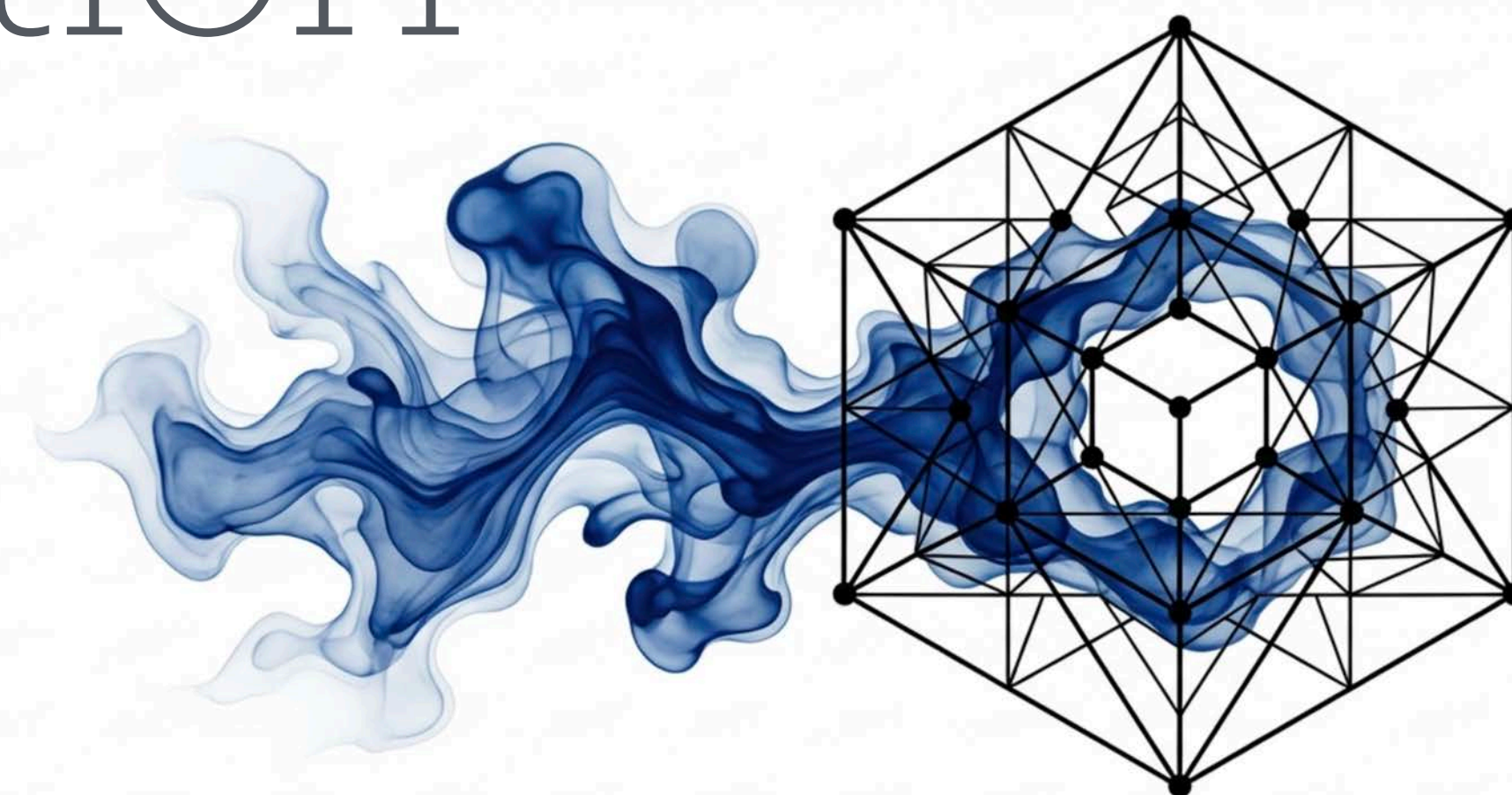


KG and LLM Integration

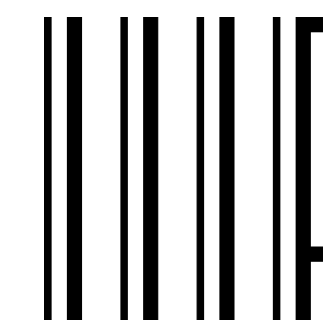
From theory to practice, part 2

Gianmaria Silvello

<https://www.dei.unipd.it/~silvello/>

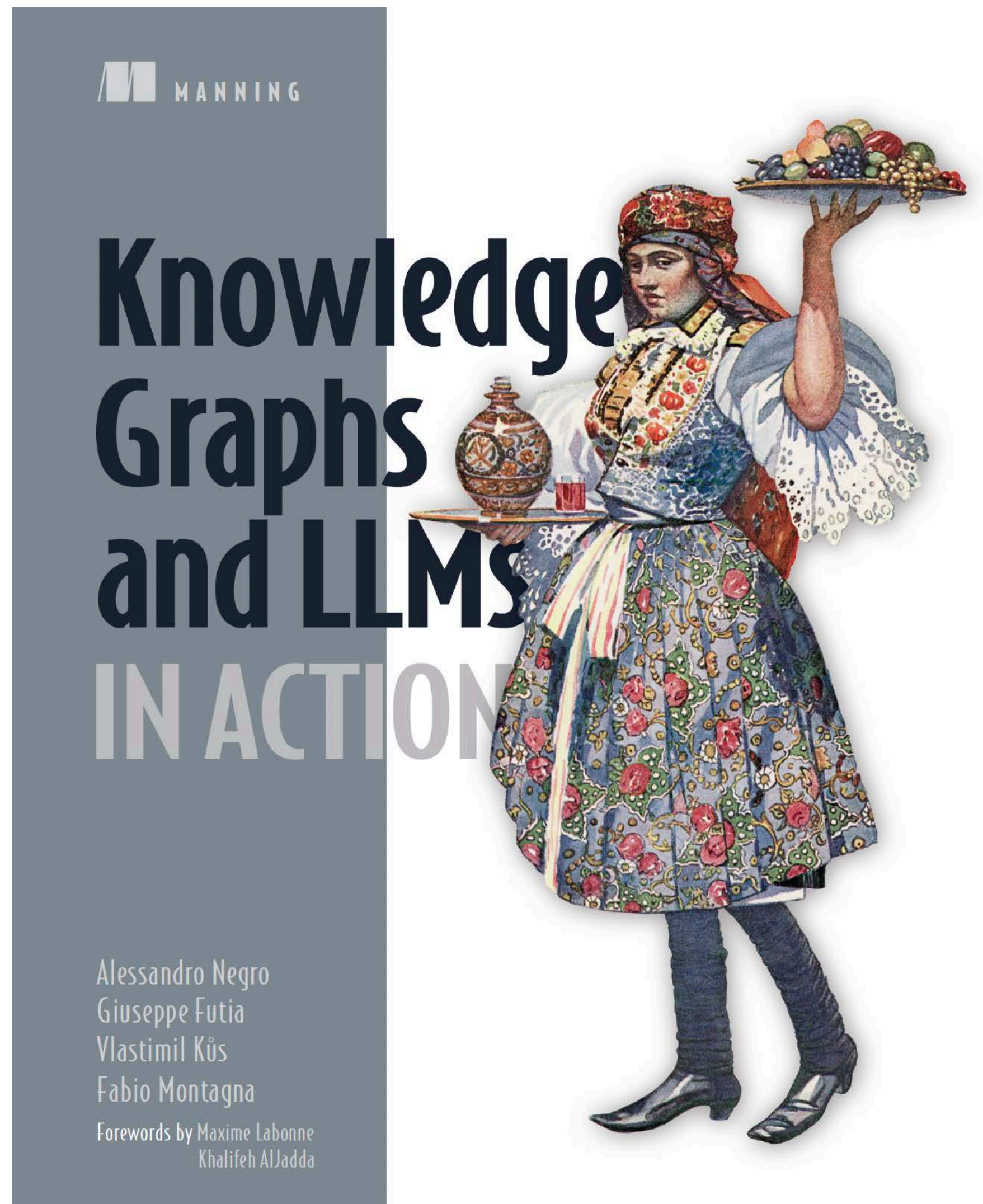


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Intelligent Interactive
Information Access
Hub

This book is the main source of reference for these slides

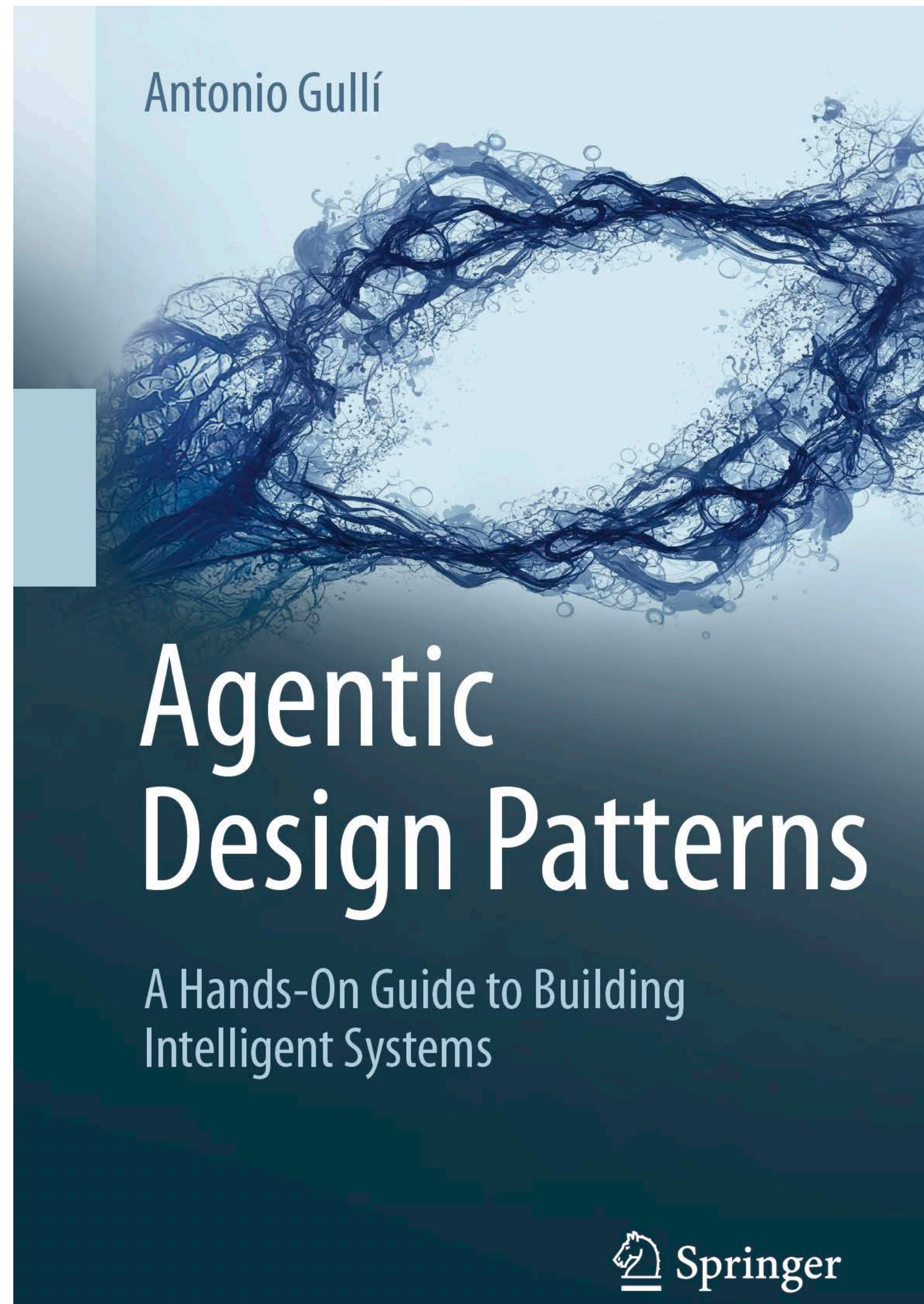


Please refer to the book and the related GitHub Repository for more in depth information and examples

I re-used some examples and some code from the book.

<https://www.manning.com/books/knowledge-graphs-and-llms-in-action>

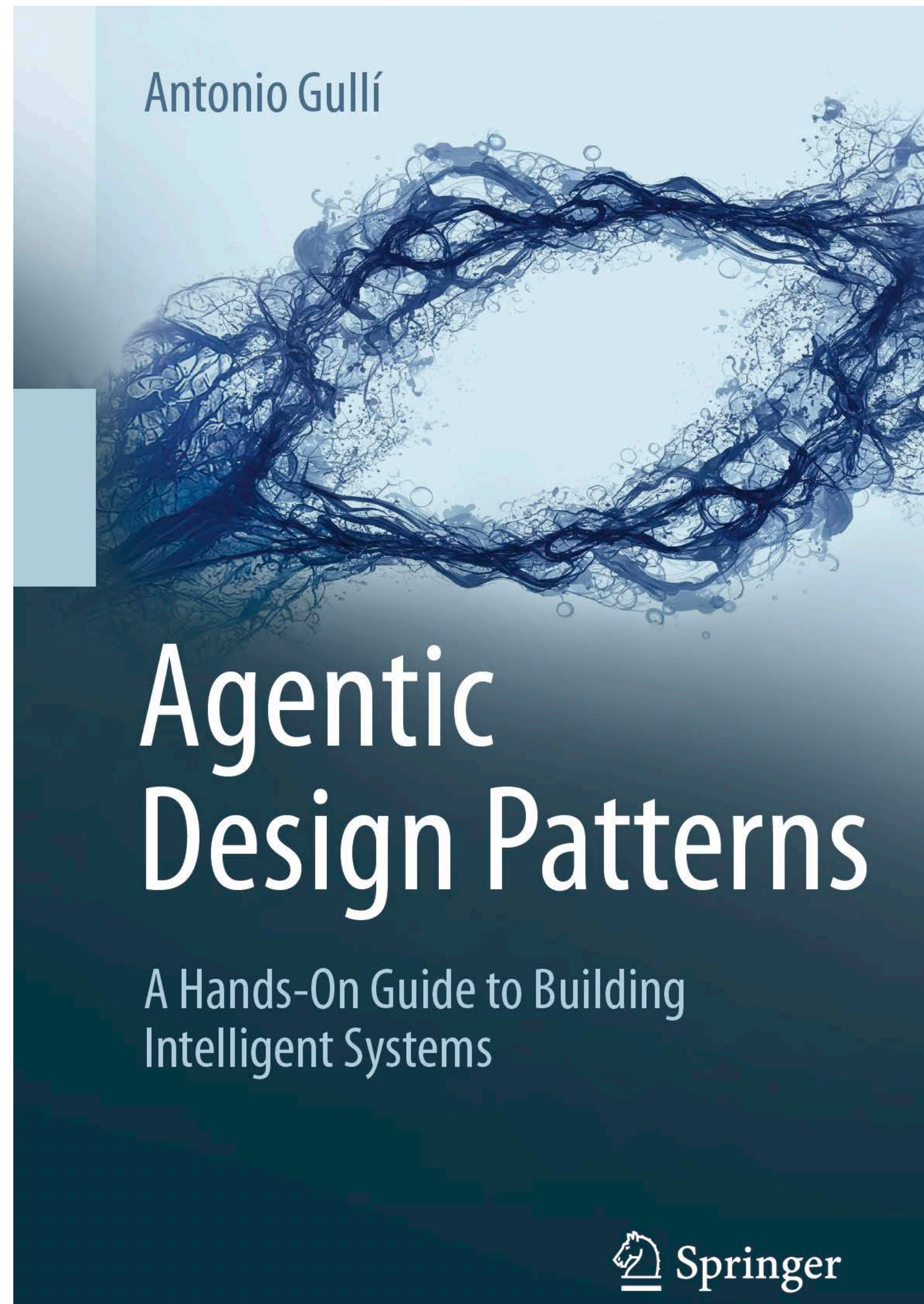
I also considered this other book



Please refer to the book and the related GitHub Repository for more in depth information and examples

https://github.com/sarwarbeing-ai/Agentic_Design_Patterns/blob/main/Agentic_Design_Patterns.pdf

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https://github.com/sarwarbeing-ai/Agentic_Design_Patterns/blob/main/Agentic_Design_Patterns.pdf

Intelligent systems

... and why KGs are the necessary backbone

Learning goals

What you'd expected to know after this deck

- Integration of Kgs and Llms
- Intelligent autonomous systems and intelligent advisory systems
- Deductive and Inductive reasoning
- Bottom-up KG construction
- CRISM-DM KG construction

Intelligent System

Intelligent systems connect users to AI and ML to achieve meaningful objectives.

An intelligent system is one in which intelligence evolves and improves over time, particularly when it improves by watching how users interact with the system.

Hulten, Geoff. 2018. Building Intelligent Systems: A Guide to Machine Learning Engineering. Apress.

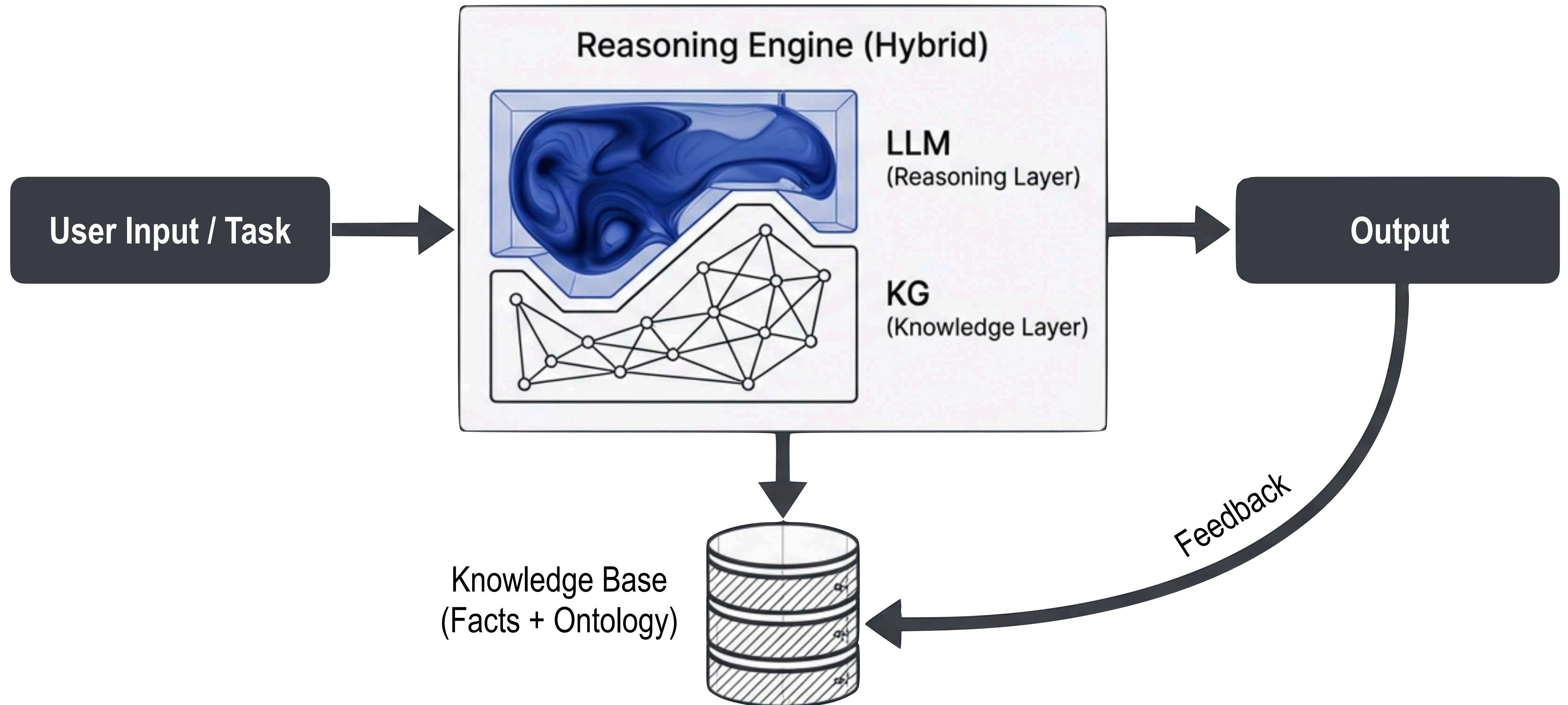
Intelligent systems

The hybrid approach

- We need a mental model for trustworthy intelligent systems
- Two core building blocks:
 - **Knowledge representation:** the “language” that AI systems use to structure, encode, and communicate information about the domain they are modelling
 - **Reasoning:** the cognitive process of analyzing information, applying rules, and drawing conclusions based on evidence or premises. We mainly consider *deductive* and *inductive* reasoning.
- We'll focus on Intelligent Advisor Systems (IASs) (support humans, don't replace them)

Architecture of a hybrid intelligent system

Intelligence is fundamentally about acquiring and applying knowledge



Designing an intelligent system

From task → architecture

- Example: “autonomous medical diagnostic assistant”
 - Must recommend a sequence of actions (tests, queries, therapies)
- Combines:
 - Established domain knowledge (ontologies, literature, structured sources)
 - Context (patient specifics, constraints, preferences)
- Key idea: algorithms form pipelines; outputs become inputs to other components

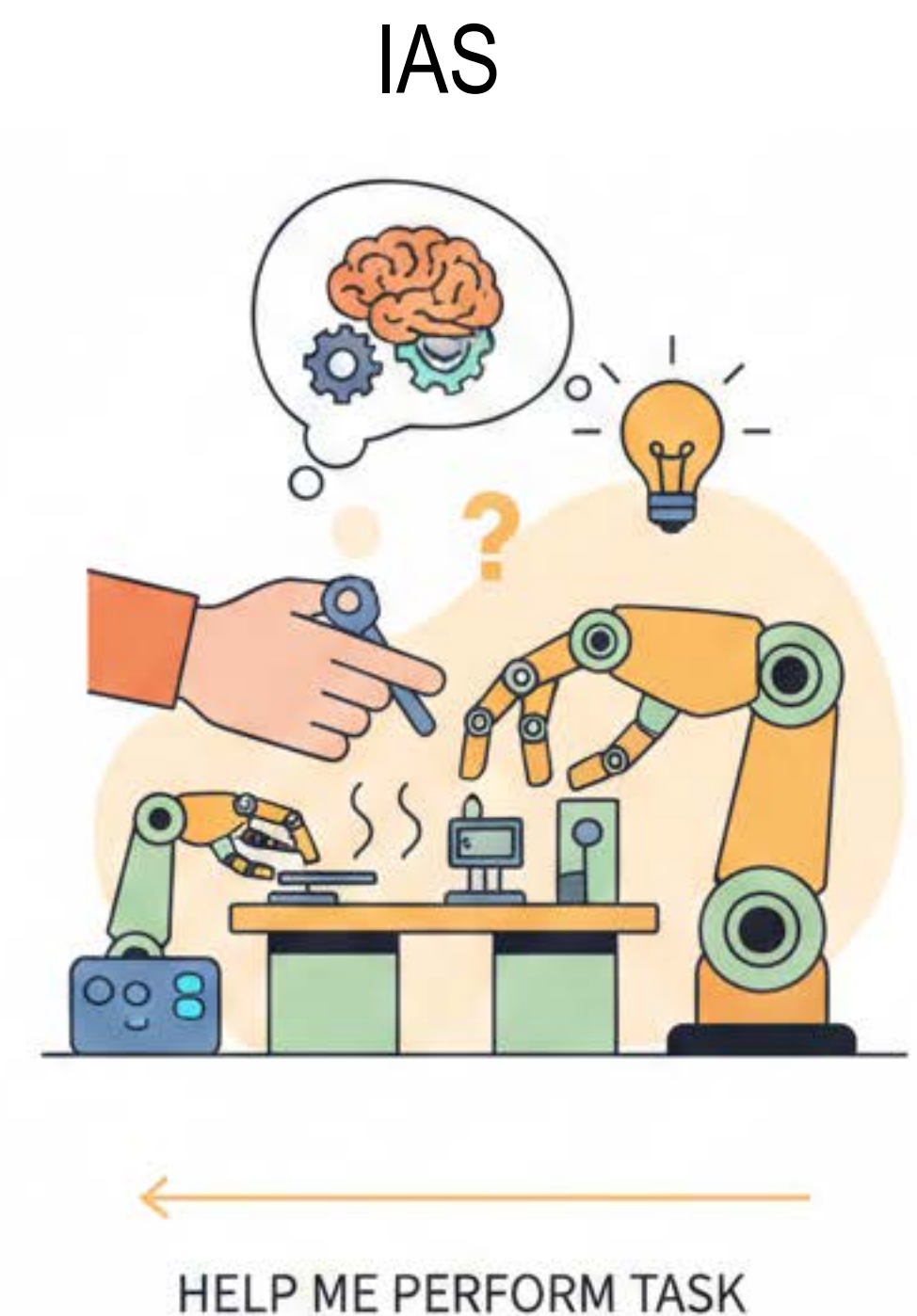
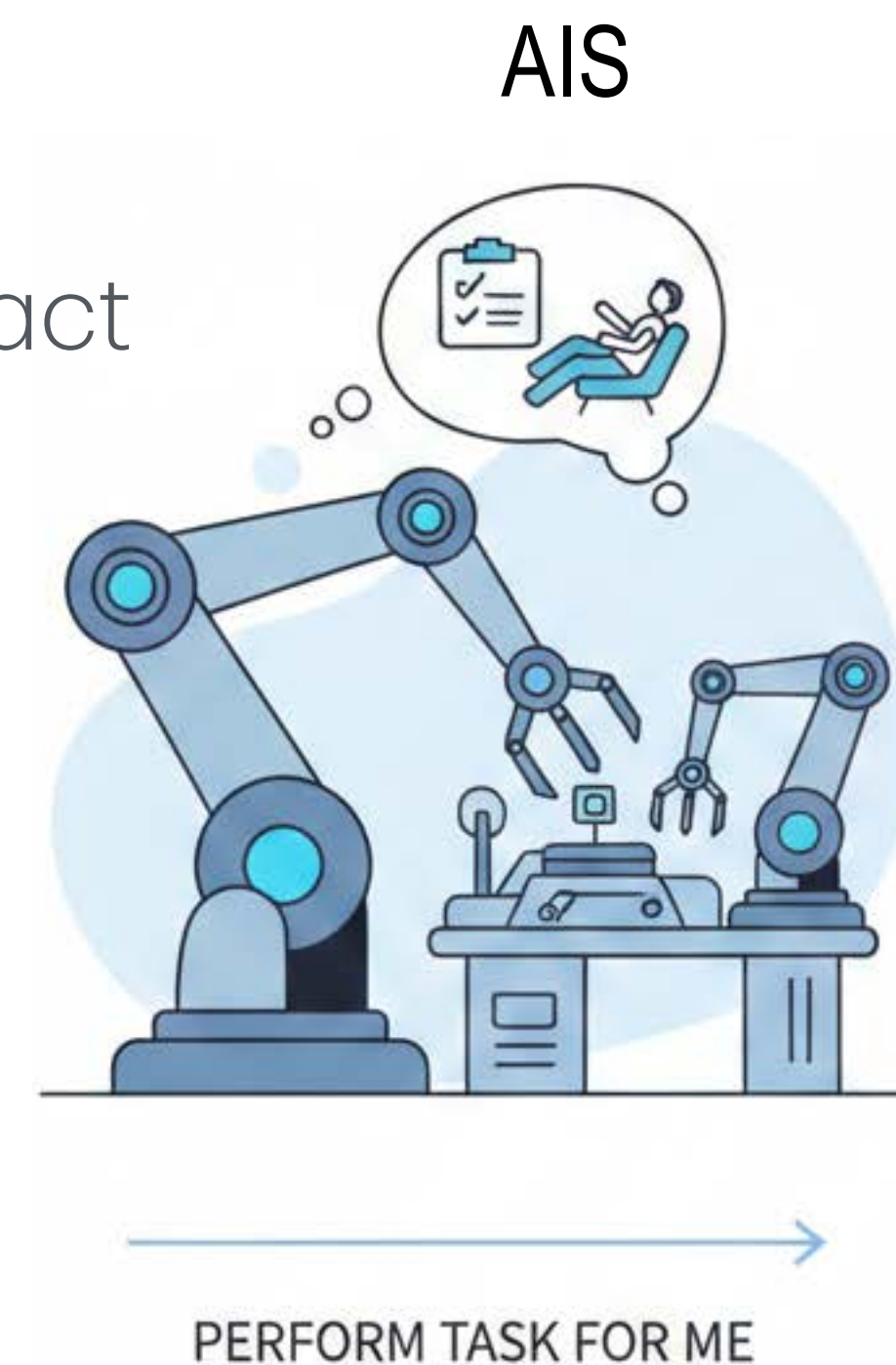
Two categories of intelligent systems

- **Autonomous Information Systems (AIS)**

- act on behalf of users (full automation, real-time decisions)

- **Information Advisor systems (IAS):**

- recommend and explain; humans decide and act
- Decision support
- Context-aware recommendations
- Interactive Q&A + explanations



Characteristics that drive design

What makes a system “intelligent”

- Meaningful objective (clear user value)
 - Define a clear, achievable goal that matters to users. This objective guides every design and implementation decision.
- Intelligent experience
 - Deliver outputs in a way that helps users reach the desired outcome. The interface should adapt to predictions, maximize upside when correct, reduce harm when wrong, and capture both implicit and explicit feedback.
- Knowledge creation and update
 - Build, maintain, and reason over knowledge continuously.
- Orchestration (multiple tools/algorithms working together safely)
 - Coordinate multiple models, algorithms, and tools in a reliable workflow.
Manage data ingestion, risk controls, and quality assurance across the system lifecycle.

Two very different knowledge acquisition patterns

- **KGs**

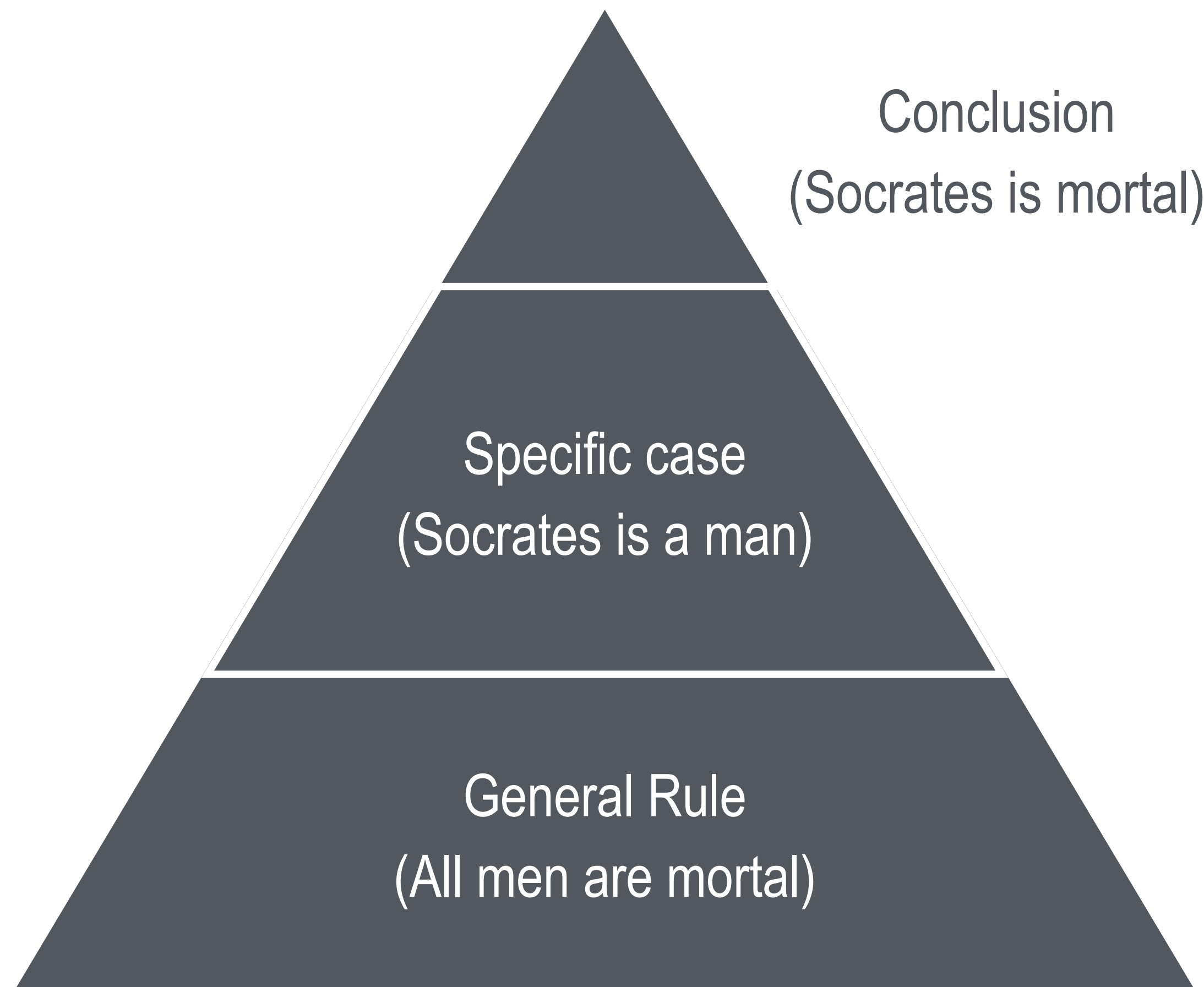
- Convert data → explicit graph (entities/relationships/properties)
- Domain experts guide semantics/schema/validation
- Human-interpretable + editable

- **LLMs**

- Learn from large-scale text via training → parameters/embeddings
- Knowledge is implicit (hard to inspect, hard to directly edit)
- Domain experts mainly influence data selection + evaluation (and sometimes fine-tuning)

Two core reasoning modes

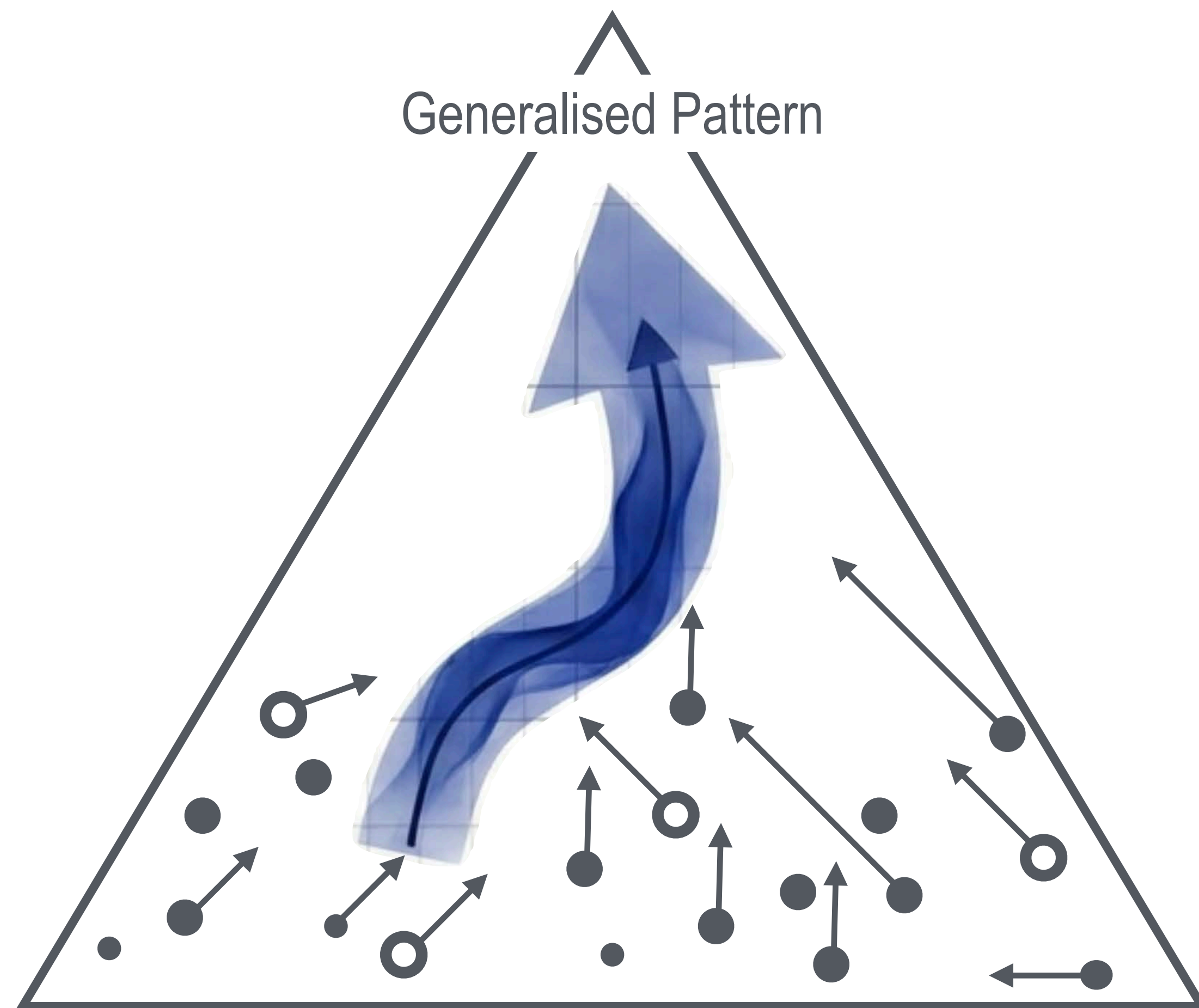
Deductive reasoning: the certainty of logics



- **Definition:** General -> specific
- **The KG role:** KGs excel in encoding explicit rules and facts
- **Constraint:** Requires a complete and accurate ontology/KB

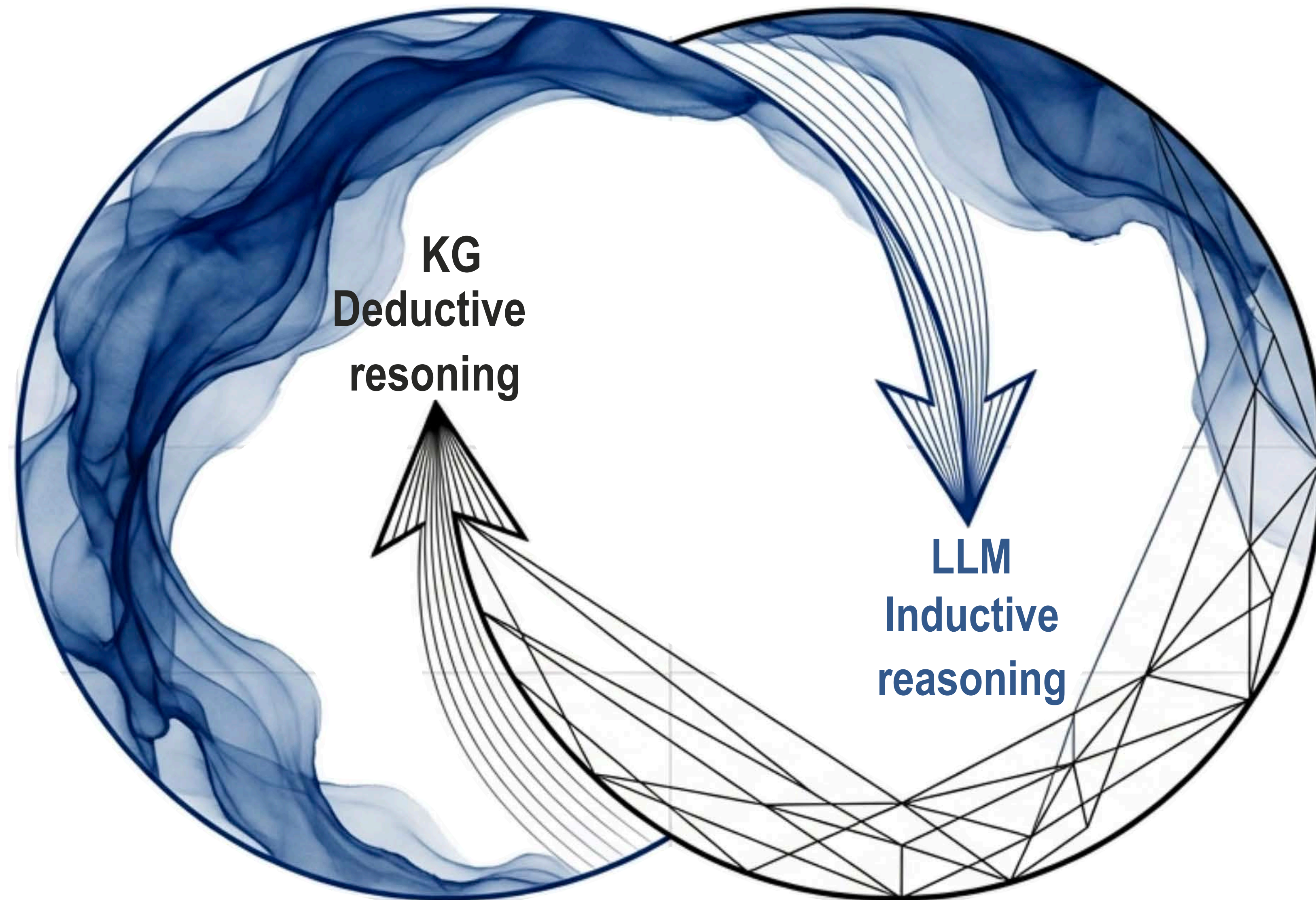
Two core reasoning modes

Inductive reasoning: managing uncertainty

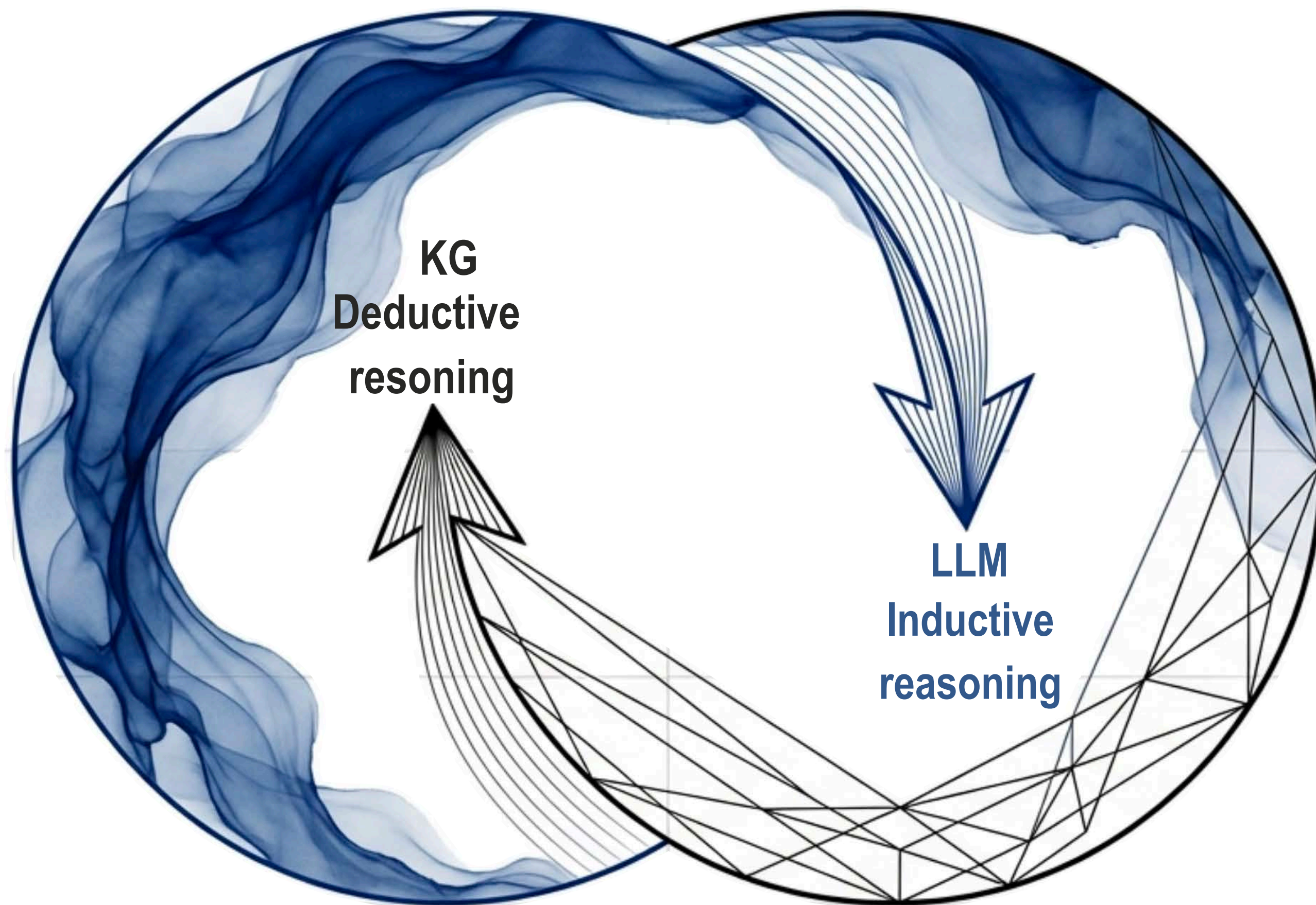


- **Definition:** Observation -> generalisation
- **The LLM role:** LLMs excel in pattern recognition, handling ambiguity, and fuzzy matching
- **Synergy:** LLMs abstract patterns from KGs to bridge gaps where explicit links are missing

We need to integrate two worlds



We need to integrate two worlds



Real IASs mix reasoning types

Deductive engine limitation: needs a highly complete knowledge base (rare in real life)

Inductive/ML helps by:

- Expanding knowledge via learned relationships/ontologies
- Handling uncertainty via prediction/generalization

LLMs contribute as a probabilistic layer that can interpret ambiguity and propose prioritized next steps

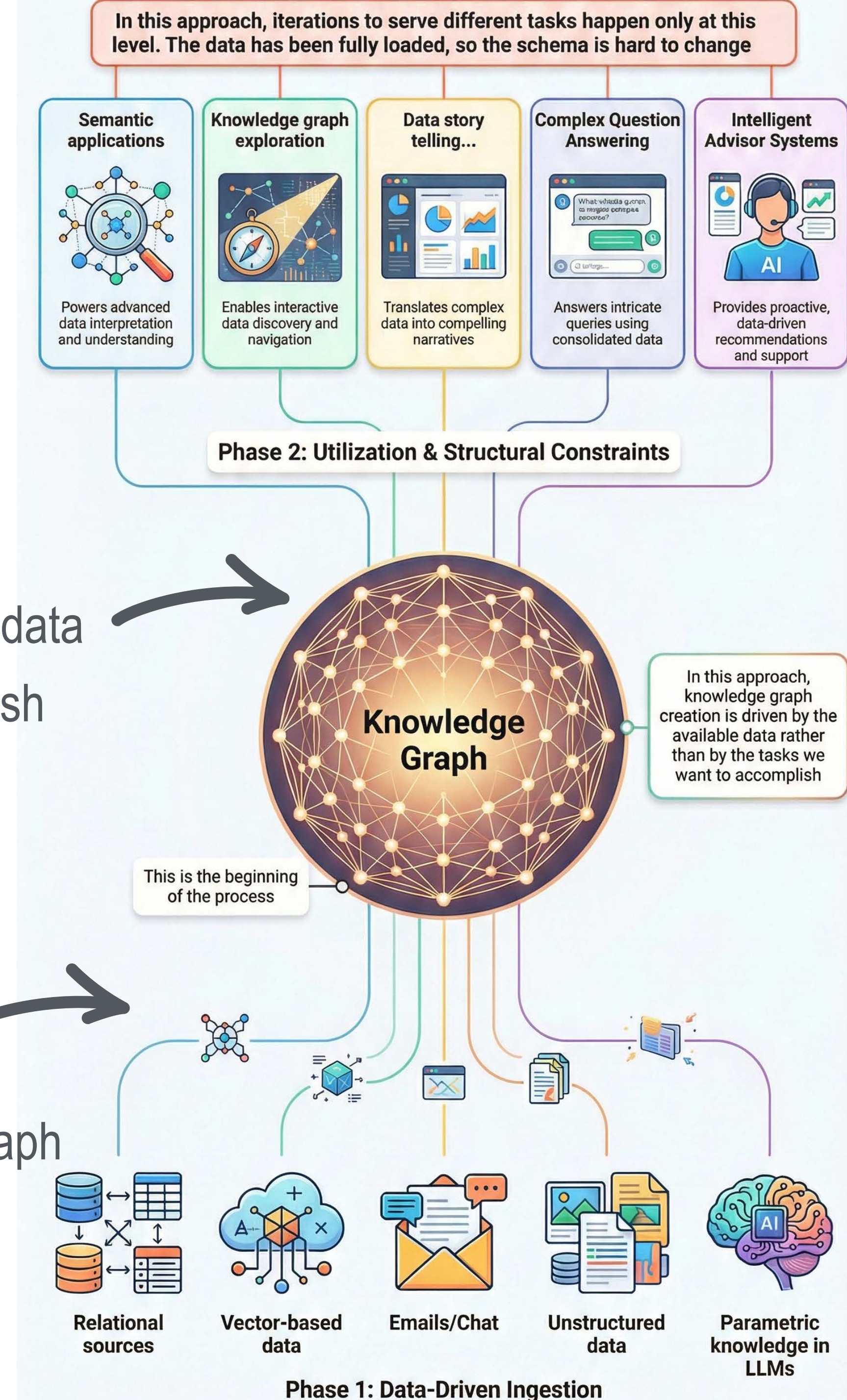
KG approach to IASs: bottom-up vs purpose-driven

Bottom-up: import everything → unify later → hope value emerges

Often too much normalization and too little task focus

KG creation is driven by the data
NOT by the task to accomplish

Data sources identification
Definition of the KG schema
Data import and convert into graph



KG approach to IASs: the CRISM-DM model

Wirth, Rüdiger, and Jochen Hipp. 2000. "CRISP-DM: Towards a Standard Process Model for Data Mining." In Proceedings of the Fourth International Conference on the Practical Application of Knowledge Discovery and Data Mining, 29–39. ACM.

KG creation is driven by the data
NOT by the task to accomplish

